# Mobile Application Development – CA1: Review 1
## Idea & Proposal Review

*Milestone: Problem definition, concept, and project scope*

| Section | Details |
|---|---|
| **Group Number** | *6* |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram Manohar (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | Pizza Delivery App |
| **App Name** | PiePoint |
| **Problem Statement** | Traditional food-ordering applications often provide limited customization and rely on static selection interfaces. Users may find it difficult to visualize their customized pizza before ordering. PiePoint aims to address this by providing an interactive pizza-building experience where users can visually see their pizza evolve as they select different ingredients, while ensuring simple navigation, responsive design, and smooth UI interactions. |
| **Abstract** | PiePoint is a UI/UX-focused Android pizza ordering application that provides an interactive and visually engaging pizza customization experience. Users can browse pizzas, build a custom pizza from scratch by selecting crusts, sauces, cheeses, and toppings, and add their creations to the cart. Smooth animations, interactive components, intuitive navigation, and responsive design are used to create an engaging and user-friendly experience. |
| **Target Users** | Gen public |
| **Planned Core Features** | • AI Pizza Builder – Generate personalized pizza suggestions based on the user's preferences and allow them to customize the generated pizza.<br>• Interactive Custom Pizza Builder – Build a pizza from scratch by selecting crust, sauce, cheese, size, and toppings with real-time visual updates.<br>• Dynamic Animations & Interactions – Smooth animations for ingredient selection, pizza building, price changes, navigation, and cart actions.<br>• Cart & Checkout – Review customized pizzas, modify quantities, and complete the ordering process.<br>• Order Management – View current and previous orders with order details and status tracking. |
| **App Icon** |  |

**Remarks:**

Create pizza builder section where the user can custom order their pizza according to their preferences.

Devnath P (25MCAR0090):

Rohan Ram Manohar (25MCAR0114):

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# Mobile Application Development – CA1: Review 2
## UI/UX Design Review

*Milestone: Wireframes, mockups, and visual design language*

| Section | Details |
|---|---|
| **Group Number** | 6 |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram Manohar (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Screens Designed** | *6 (Home, Pizza builder, Pizza Screen, Cart and transaction, Profile, Order history)* |
| **Wireframe / Mockup Tool** |   |
| **Colour Palette & Typography** | • ***Brand:*** *Orange*<br>• ***Neutral:*** *White*<br>• ***Typography:*** *Navy*<br>• ***Gradients:*** *Premium* |
| **Design Consistency Notes** | • ***Glassmorphism:*** *Frosted*<br>• ***Borders:*** *Glowing*<br>• ***Corners:*** *Rounded*<br>• ***Feedback:*** *Tactile* |
| **Layout & Visual Balance** | • ***Grid:*** *8dp*<br>• ***Spacing:*** *Consistent*<br>• ***Graphics:*** *Floating*<br>• ***Hierarchy:*** *Clear* |
| **Add-ons as per Remarks** | *Created Pizza builder section where the user can create a custom pizza of their choice.* |

Evaluation – UI Design & Visual Appeal

**Remarks:**

Add an option to increase and decrease the amount of cheese which can be added.

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Prajwal B S (25MCAR0228):

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# Mobile Application Development – CA1: Review 3

## Development & Responsiveness Review

*Milestone: Core build progress and responsive layout implementation*

| Section | Details |
|---|---|
| **Group Number** | 6 |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram Manohar (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Features Implemented** | 1.Interactive 3D Pizza Builder: Step-by-step custom wizard (Crust $\rightarrow$ Sauce $\rightarrow$ Cheese $\rightarrow$ Toppings $\rightarrow$ Size $\rightarrow$ Review) with a real-time animated overlay preview and a 3D rotation fly-to-cart effect.<br>2. Sleek Custom Bottom Navbar: Fluid notch design that matches tab shifts and morphs accent colors per category dynamically.<br>3. Cinematic Order Confirmation: Real-time delivery stage timeline showing a 3D pizza box packing, cargo loading, and the delivery van driving away with vibrating engine animations.<br>4. Cart & Checkout Flow: Full state management handling item quantities, price modifiers, and order placements. |
| **Tech Stack Used** | • Core Language: Kotlin<br>• UI Toolkit: Jetpack Compose (Material Design 3 & custom Canvas paths)<br>• Navigation: Jetpack Navigation Compose<br>• Architecture: MVVM (Model-View-ViewModel) with StateFlow and Coroutines<br>• Image Loading: Coil-Compose<br>• Data Serialization: Kotlinx Serialization JSON<br>• Target Environment: Android SDK 35 (Compile 35, Min SDK 26) |
| **Devices / Screen Sizes Tested** | • Mobile Phones: Pixel 8 Pro (high density, tall ratio), Pixel 7a (medium/standard mobile layout)<br>• Density Scales: Tested MDPI up to XXXHDPI screen densities to verify launcher icons and vector assets rendering quality. |
| **Responsive Design Approach** | • Dynamic Layout Measurement: The custom notched navbar utilizes onSizeChanged to dynamically measure screen width in pixels and calculate proportional tab sizes in DP on the fly.<br>• Adaptive Scroll Containers: Columns and detail views utilize verticalScroll() and lazy containers (LazyRow/LazyColumn) to prevent UI overflow on smaller height devices.<br>• Proportional Sizing: Layout coordinates, sizes, and padding are built with responsive weight modifiers (Modifier.weight()) and relative aspect ratios instead of hardcoded coordinates.<br>• Density Independence: System dimensions use strictly dp and sp measurements, honoring user-defined device scaling. |
| **Known Issues / Bugs** | • Visual Density on Ultra-Narrow Devices: On screens narrower than 320dp, the 5-tab floating bottom navigation labels are closely spaced, though fully legible. |
| **Add-ons as per Remarks** | Added the option to increase and decrease toppings as user users wants. |

**Evaluation – Responsiveness & Interactivity:**

**Remarks:**

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**Signature of Members :**

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# Mobile Application Development – CA1: Review 4
# UX, Navigation & Usability Testing

*Milestone: User flow, interactivity, and usability feedback*

| Section | Details |
|---|---|
| **Group Number** | 6 |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram Manohar (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Navigation Flow Summary** | *Users navigate through a bottom navigation bar (Menu, Create, Offers, Orders, Profile) leading to item discovery or the multi-step custom pizza builder. From there, items route seamlessly to the Cart, advance to Checkout, and complete at OrderConfirmation featuring a cinematic delivery handoff with direct paths to OrderHistory.* |
| **Interactive Components Used** | *The interface leverages tactile PizzaCard elements with 0.96× bouncy spring physics, pill-shaped QuantitySelector controls, and a swipeable hero HorizontalPager. It also integrates dynamic layer toggles in the PizzaBuilder alongside animated GradientButton CTAs and badge-updating floating cart icons* |
| **Usability Testing Method** | *Task-based scenario testing was conducted where users performed core flows—such as assembling a custom pizza from crust to toppings and executing checkout—monitored on physical Android devices. This was paired with heuristic walkthroughs focusing on tap target comfort (8dp grid system) and animation latency.* |
| **Usability Feedback Received** | *Participants commended the instant visual price updates, tactile button feedback, and engaging 3D "Fly-to-Cart" and delivery van departure animations. Feedback suggested streamlining the step-by-step custom builder to reduce clicks when switching previously selected ingredients.* |
| **Navigation Improvements Planned** | *Planned enhancements include adding interactive step breadcrumbs to the pizza builder for direct, non-linear ingredient editing without sequential back-tapping. Additionally, one-tap "Reorder" shortcuts will be embedded into OrderHistory and HomeScreen to bypass repetitive cart rebuilding.* |
| **Add-ons as per Remarks** | *Nil* |

**Evaluation – User Experience & Navigation**

**Remarks:**

Devnath P (25MCAR0090):
Rohan Ram Manohar (25MCAR0114):
Prajwal B S (25MCAR0228):

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# Mobile Application Development – CA1: Review 5

## Final Presentation & Deployment Review

*Milestone: Completed app, deployment, and final demo*

| Section | Details |
|---|---|
| **Group Number** | *6* |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram Manohar (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Deployment Details** | *Built as a native Android application using Gradle and Jetpack Compose, packaged into release APK/AAB bundles via ./gradlew assembleRelease. The app targets modern Android API levels (API 26+) with ProGuard/R8 minification enabled for streamlined distribution via Google Play Console internal testing tracks.* |
| **Final Feature Summary** | *PiePoint delivers a complete ordering ecosystem featuring a dynamic catalog with auto-sliding hero banners, an interactive custom pizza builder with real-time layer rendering and price calculations, and persistent cart management. The experience concludes with a checkout flow and an immersive, 3-stage cinematic order packing and delivery dispatch sequence tied to past order tracking.* |
| **Demo Notes** | *Begin on the Home screen to showcase category filtering and bouncy card touch interactions, then enter the "Create" builder to demonstrate real-time topping stacking and the 3D "Fly-to-Cart" spiral effect. Conclude by placing an order to highlight the automated box-closing and delivery van departure animations before viewing the updated status in Order History.* |
| **Project Report Submitted** | *https://github.com/devnath-p/PiePoint* |
| **App Icon** |  |
| **Add-ons as per Remarks** | *NIL* |

**Final Evaluation – CA1 Rubric Total 15 Marks**

**Remarks:**

Devnath P (25MCAR0090):
Rohan Ram Manohar (25MCAR0114):
Prajwal B S (25MCAR0228):

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# Mobile Application Development – CA3: Review 1

## Idea & AI Use-Case Proposal

*Milestone: Problem definition, AI/ML use case, and project scope*

| Section | Details |
|---|---|
| **Group Number** | *6* |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Proposed AI/ML Feature** | *Intelligent decision support / recommendation: an AI pizza builder and "Ask Pie" chatbot that turn free-text requests into valid custom pizzas.* |
| **Why AI is Needed** | *Replaces tedious multi-step selection (crust, sauce, cheese, toppings, size) by converting natural-language cravings, diet and budget into a ready pizza in seconds.* |
| **Target Users** | *Pizza customers who want menu pizzas, custom pizzas, or quick AI-suggested pizzas matching their taste and budget.* |
| **Add-ons as per Remarks** | *Nil* |

**Remarks:**

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# Mobile Application Development – CA3: Review 2

## AI/ML Model & Data Review

*Milestone: Dataset preparation, model selection, and training approach*

| Section | Details |
|---|---|
| **Group Number** | *6* |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Dataset Used** | *None (pre-trained LLM). Menu data (menu.json, 33 items) from the app is given as context.* |
| **Data Preprocessing** | *Menu injected into prompt; prompt checks (non-empty, ≤500 chars); JSON-only output; IDs validated, price computed server-side.* |
| **Model / Algorithm Chosen** | *Pre-trained LLM qwen/qwen3.8-27b via Groq Cloud API; free and fast.* |
| **Training & Validation Approach** | *No training. Tested with 20 prompts (normal, edge, off-topic) on /create-pizza.* |
| **Model Performance (so far)** | *19/20 successful (1 empty prompt rejected); 95% valid JSON first try; 90% valid IDs; avg 4.41 s.* |
| **Integration Plan** | *Cloud API: Android (Retrofit) → Node.js/Express backend → Groq API.* |
| **Add-ons as per Remarks** | *Nil* |

**Evaluation – AI Integration & Performance (partial)**

**Remarks:**

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# Mobile Application Development – CA3: Review 3

## App Development & Functionality Review

*Milestone: Core app build and functional integration of the AI feature*

| Section | Details |
|---|---|
| **Group Number** | *6* |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Features Implemented** | *Menu, pizza details, 5-step custom builder, AI builder + Ask Pie chat, cart, checkout, order history, profile.* |
| **Tech Stack Used** | *Kotlin, Jetpack Compose, Retrofit/OkHttp; Node.js + Express backend; Groq API.* |
| **AI Feature Functionality** | *Text/chip in Ask Pie → /chat → backend adds menu, calls Groq → validates IDs, prices → pizza card (Add to Cart / Customize).* |
| **Error Handling / Edge Cases** | *Empty prompt → 400; rate limit → friendly message; bad JSON → 1 retry; unknown items → defaults; offline → sample data.* |
| **Known Issues / Bugs** | *Currency mismatch (₹ backend, $ app); AI prefill lost on process death;* |
| **Add-ons as per Remarks** | *Nil* |

**Evaluation – Application Functionality**

**Remarks:**

Fix the limit on toppings that can be added

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# Mobile Application Development – CA3: Review 4

## UI/UX & AI Output Presentation Review

*Milestone: Interface design and clarity of AI-generated results*

| Section | Details |
|---|---|
| **Group Number** | *6* |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Screens Designed** | *Splash, Home, Pizza Detail, Create, Ask Pie, Cart, Checkout, Order Confirmation, Order History, Profile.* |
| **Navigation Flow** | *5-tab bottom bar (Create, Ask Pie, Menu, Orders, Profile); Ask Pie opens AI chat; pizza cards go to cart or builder.* |
| **AI Output Presentation** | *Chat bubbles, typing indicator, pizza card with ingredients and ₹ price, Add to Cart / Customize buttons.* |
| **Interactive Components Used** | *Buttons, chips, text fields, cards, lists, pager, progress indicator, animated bottom bar, toast.* |
| **Responsiveness Check** | *Pixel 10a (emulator); vivo X200 FE (physical device).* |
| **Add-ons as per Remarks** | *Fixed the limit on toppings that can be added* |

**Evaluation – UI/UX Design**

**Remarks:**

*Fix UI bug where the textbox was not displayed properly*

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# Mobile Application Development – CA3: Review 5

## Final Deployment & Demo Review

*Milestone: Deployed application, final AI performance, and live demo*

| Section | Details |
|---|---|
| **Group Number** | *6* |
| **Team Members** | *Devnath P (25MCAR0090), Rohan Ram (25MCAR0114), Prajwal B S (25MCAR0228)* |
| **Title** | *Pizza Delivery App* |
| **App Name** | *PiePoint* |
| **Deployment Details** | *https://github.com/devnath-p/PiePoint* |
| **Final AI Model Performance** | *Same as Review 2 (19/20 successful, 95% valid JSON, avg 4.41 s).* |
| **Final Feature Summary** | *Menu, custom builder, AI builder, Ask Pie chatbot, cart, checkout, order history, profile.* |
| **Demo Notes** | *Browse menu; build custom pizza; ask Ask Pie for "spicy veg pizza"; Customize; add to cart, check out, view history.* |
| **Project Report Submitted** | *https://drive.google.com/file/d/1eJzpAu4Kc4NjYY8IoL27O6EiBuvRHNth/view?usp=sharing* |
| **Add-ons as per Remarks** | *Fixed UI bug where the textbox was not displayed properly* |

**Final Evaluation – CA3 Rubric (Total 15 Marks)**

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